

Practical No : 4

Tutorial Title: Write a program in C++ to perform the following operations on complex numbers: Add, Subtract, Multiply, Divide and Complex Conjugate. Design a class for complex number representation and the operations to be performed.

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Program:

```
#include <iostream>
using namespace std;
class Complex {
    float real, imag;
public:
    void read() {
        cin >> real >> imag; }
    void display() {
        if (imag >= 0)
            cout << real << " + " << imag << "i";
        else
            cout << real << " - " << -imag << "i"; }
    Complex add(Complex c) {
        Complex t;
        t.real = real + c.real;
        t.imag = imag + c.imag;
        return t; }
    Complex subtract(Complex c) {
        Complex t;
        t.real = real - c.real;
        t.imag = imag - c.imag;
        return t; }
    Complex multiply(Complex c) {
        Complex t;
        t.real = real * c.real - imag * c.imag;
        t.imag = real * c.imag + imag * c.real;
        return t; }
    Complex divide(Complex c) {
```

```

    Complex t;
    float d = c.real * c.real + c.imag * c.imag;
    t.real = (real * c.real + imag * c.imag) / d;
    t.imag =(imag * c.real - real * c.imag) / d;
    return t; }
Complex conjugate() {
    Complex t;
    t.real = real;
    t.imag = -imag;
    return t;}    };
int main() {
    Complex c1, c2;
    Complex add, sub, mul, div, con;
    cout << "Enter first complex number (real imaginary): ";
    c1.read();
    cout << "Enter second complex number (real imaginary): ";
    c2.read();
    add = c1.add(c2);
    sub = c1.subtract(c2);
    mul = c1.multiply(c2);
    div = c1.divide(c2);
    con = c1.conjugate();
    cout << "\nAddition = ";
    add.display();
    cout << "\nSubtraction = ";
    sub.display();
    cout << "\nMultiplication = ";
    mul.display();
    cout << "\nDivision = ";
    div.display();
    cout << "\nComplex Conjugate = ";
    con.display();
    return 0;}

```

Output:

```
#include <iostream>
using namespace std;
class Complex {
    float real, imag;
public:
    void read() {
        cin >> real >> imag; }
    void display() {
        if (imag >= 0)
            cout << real << " + " << imag << "i";
        else
            cout << real << " - " << -imag << "i"; }
    Complex add(Complex c) {
        Complex t;
        t.real = real + c.real;
        t.imag = imag + c.imag;
        return t; }
    Complex subtract(Complex c) {
        Complex t;
        t.real = real - c.real;
        t.imag = imag - c.imag;
        return t; }
    Complex multiply(Complex c) {
        Complex t;
        t.real = real * c.real - imag * c.imag;
        t.imag = real * c.imag + imag * c.real;
        return t; }
    Complex divide(Complex c) {
        Complex t;
        float d = c.real * c.real + c.imag * c.imag;
        t.real = (real * c.real + imag * c.imag) / d;
        t.imag = (imag * c.real - real * c.imag) / d;
        return t; }
    Complex conjugate() {
        Complex t;
        t.real = real;
        t.imag = -imag;
        return t; }
};

int main() {
    Complex c1, c2;
    c1.read();
    c2.read();
    Complex add = c1.add(c2);
    Complex sub = c1.subtract(c2);
    Complex mul = c1.multiply(c2);
    Complex div = c1.divide(c2);
    Complex con = c1.conjugate();
    con.display();
    add.display();
    sub.display();
    mul.display();
    div.display();
    return 0;
}
```

Enter first complex number (real imaginary): 3 4
Enter second complex number (real imaginary): 2 6

Addition = 5 + 10i
Subtraction = 1 - 2i
Multiplication = -18 + 26i
Division = 0.75 - 0.25i
Complex Conjugate = 3 - 4i

=== Code Execution Successful ===

```
main.cpp Run Output
1: return t; }
2: Complex divide(Complex c) {
3:     Complex t;
4:     float d = c.real * c.real + c.imag * c.imag;
5:     t.real = (real * c.real + imag * c.imag) / d;
6:     t.imag = (imag * c.real - real * c.imag) / d;
7:     return t; }
8: Complex conjugate() {
9:     Complex t;
10:    t.real = real;
11:    t.imag = -imag;
12:    return t; }
13: int main() {
14:    Complex c1, c2;
15:    Complex add, sub, mul, div, con;
16:    cout << "Enter first complex number (real imaginary): ";
17:    c1.read();
18:    cout << "Enter second complex number (real imaginary): ";
19:    c2.read();
20:    add = c1.add(c2);
21:    sub = c1.subtract(c2);
22:    mul = c1.multiply(c2);
23:    div = c1.divide(c2);
24:    con = c1.conjugate();
25:    con.display();
26:    add.display();
27:    sub.display();
28:    mul.display();
29:    div.display();
30:    cout << "Complex Conjugate = ";
31:    con.display();
32:    return 0;
}
```

Enter first complex number (real imaginary): 3 4
Enter second complex number (real imaginary): 2 6

Addition = 5 + 10i
Subtraction = 1 - 2i
Multiplication = -18 + 26i
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Complex Conjugate = 3 - 4i

=== Code Execution Successful ===

